

? Question 1

When you run `np.spacing(1e7)`, you see `1.862645149230957e-09` as a result.

1. If you run `1e7 == 1e7+np.spacing(1e7)`, what will you see as a result? Why?
2. If you run `1e7 == 1e7+np.spacing(1e7)/3`, what will you see as a result? Why?

? Question 2

Please convert the following floating point number to IEEE754 representation (by double precision):

-10.125

? Question 3

Please convert the following IEEE754 representation to decimal (base 10) number (in double precision):

1 10000000011 10100000000.....000

? Question 4

Given the vector $X = (x_1, x_2, x_3)$, please explain and distinguish L1 and L2 norms.

? Question 5

What does "ill-conditioned matrix" mean? Please explain it in terms of numerical errors.

? Question 6

How do you check if a matrix has its corresponding inverse matrix? You may show it with an example.

? Question 7

The following Python code written in Ipython is given:

```
import numpy as np
np.spacing(1e15)
print(1e15 + (0.125/3)) # Output: 1000000000000000.0
```

1. Please explain what the code does.
2. Why does Python return 1e15 after the addition?

? Question 8

What is the largest number which is smaller than 7.1 in IEEE754 representation (in single precision)?

? Question 9

Please explain round-off error with an example.

? Question 10

Please write a class called `BankAccount` with the following specifications:

- Attributes: `__balance`, `__accountOwner`, `__accountType`
- `__init__()` initializes the attributes
- Mutators: `set_balance()`, `set_accountOwner()`, `set_accountType()`
- Accessors: `get_balance()`, `get_accountOwner()`, `get_accountType()`

? Question 11

Which of the following will create an object, `worker_joey`, of the `Worker` class?

- ☐ `def __init__(worker_joey):`
- ☐ `class worker_joey:`
- ☒ `worker_joey = Worker()`
- ☐ `worker_joey.Worker`

? Question 12

Which is the first line needed when creating a class named `Worker`?

- ☐ `def __init__(self):`
- ☒ `class Worker:`
- ☐ `import random`
- ☐ `def worker_pay(self):`

? Question 13

What is the relationship called in which one object is a specialized version of another object?

- ☐ parent-child
- ☐ node-to-node
- ☒ is a
- ☐ class-subclass

? Question 14

All instances of a class share the same values of the data attributes in the class.

- ☐ True
- ☒ False

? Question 15

A class can be thought of as a blueprint that can be used to create an object.

- ☒ True
- ☐ False

? Question 16

New attributes and methods may be added to a subclass.

- ☒ True
- ☐ False

? Question 17

A value-returning function is like a simple function except that when it finishes it returns a value back to the part of the program that called it.

- ☒ True
☐ False

? Question 18

A function definition specifies what a function does and causes the function to execute.

- ☐ True
☒ False

? Question 19

One reason not to use global variables is that it makes a program hard to debug.

- ☒ True
☐ False

? Question 20

What will be displayed after the following code is executed?

```
def pass_it(x, y):  
    z = x * y  
    result = get_result(z)  
    return(result)  
  
def get_result(number):  
    z = number + 2  
    return(z)  
  
num1 = 3  
num2 = 4  
answer = pass_it(num1, num2)  
print(answer)
```

- ☐ 12
☐ 9
☒ 14
☐ Nothing, this code contains a syntax error.

? Question 21

In the following function, which of the parameters are keyword-only parameters?

```
def show_values(a, b, *, c, d):  
    print(a, b, c, d)
```

- ☐ a and b
- ☒ c and d
- ☐ a, b, and *
- ☐ a, b, c, and d

? Question 22

A set of statements that belong together as a group and contribute to the function definition is known as a

- ☐ header
- ☒ block
- ☐ return
- ☐ parameter

? Question 23

A problem can normally be solved with recursion if it can be broken down into smaller problems that are identical in structure to the overall problem.

- ☒ True
- ☐ False

? Question 24

Recursive algorithms are always more concise and efficient than iterative algorithms.

- ☐ True
- ☒ False

? Question 25

Each time a function is called in a recursive solution, the system incurs overhead that is not incurred with a loop.

- ☒ True
- ☐ False

? Question 26

Which would be the base case in a recursive solution to the problem of finding the factorial of a number?

- ☒ $n = 0$
- ☐ $n = 1$
- ☐ $n > 0$
- ☐ The factorial of a number cannot be solved with recursion.

? Question 27

A problem can be solved with recursion if it can be broken down into __ problems.

- ☒ smaller
- ☐ one-line
- ☐ manageable
- ☐ modular

? Question 28

In a recursive solution, if the problem cannot be solved now, then a recursive function reduces it to a smaller but similar problem and

- ☐ exits
- ☐ returns to the main function
- ☐ returns to the calling function
- ☒ calls itself to solve the smaller problem